UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

ECE 598 PROJECT

Dynamic Object Handover: Real-Time Adaptive Human-Robot Interaction

A vision-based robotic system that autonomously grasps objects and dynamically adapts handover trajectory as humans move their hands in real-time.

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Introduction to Humanoid Robotics | December 2025

Platform

ROBOTIS AI Worker | MuJoCo

The Problem

CURRENT STATE

Static & Robotic Handovers

- Robot moves to fixed position and waits
- Human must reach to exact location
- Feels unnatural and mechanical
- Human adapts to robot, not vice versa

THIS PROJECT

Dynamic Adaptive Handover

- Robot tracks human hand in real-time
- Arm mirrors hand movement dynamically
- Gesture-triggered release mechanism
- Natural, intuitive interaction

Applications: Healthcare & Elderly Care | Service Robotics | Collaborative Manufacturing

System Architecture

1 PERCEPTION

Object Detection

RGB-D camera, HSV segmentation, 3D position via depth

2 MANIPULATION

Pick Sequence

Jacobian IK, Position, Descend, Grasp, coordinated motion

3 TRANSPORT

Safe Navigation

Lift with tilt, strafe to clear, rotate 90 deg, approach

4 HANDOVER

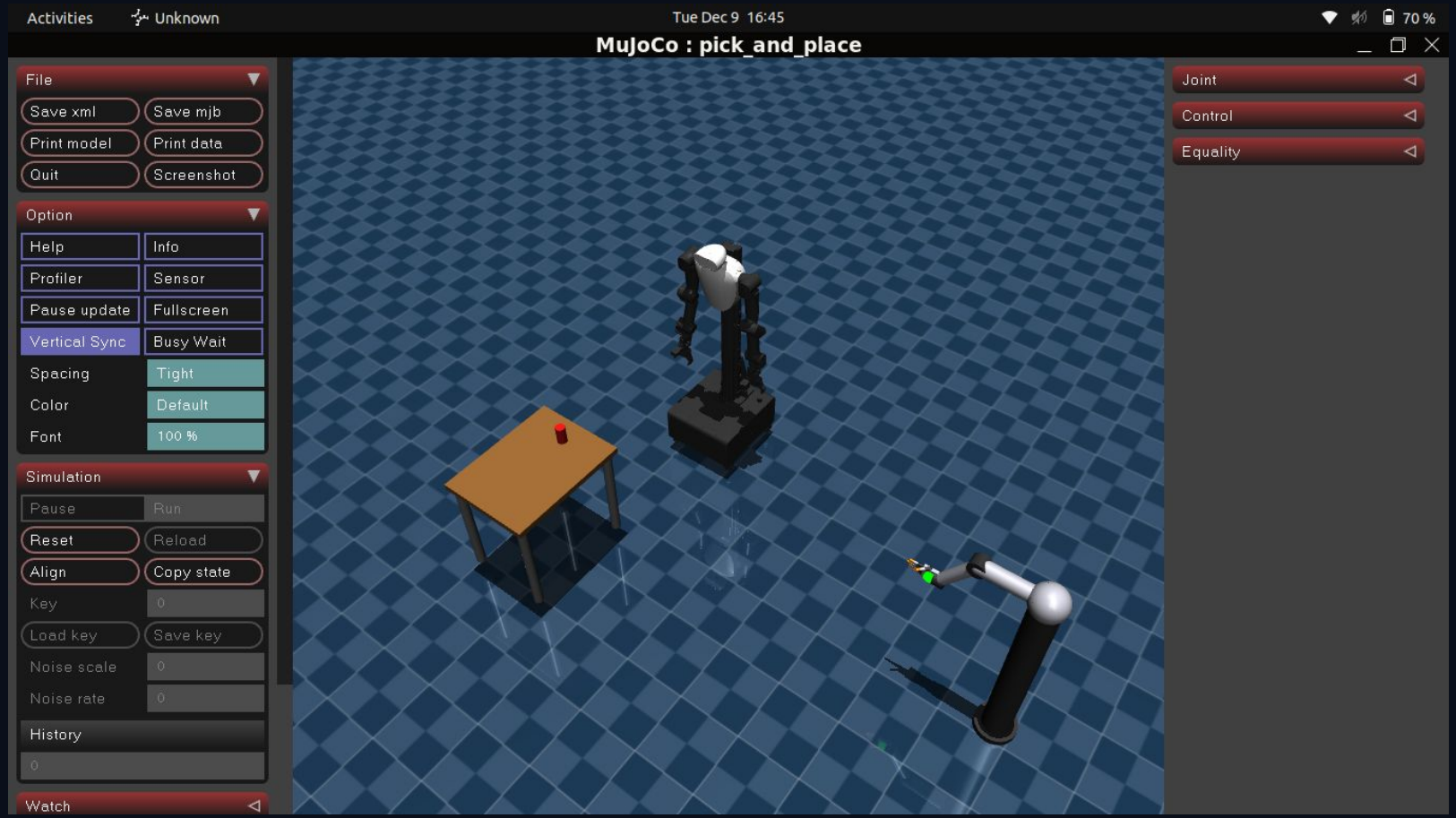
Dynamic Tracking

MediaPipe tracking, real-time mirroring, gesture release

12-STATE MACHINE

SEARCHING → APPROACHING → POSITIONING → DESCENDING → GRASPING → LIFTING → TILTING → STRAFING → ROTATING → TRACKING → RELEASING → COMPLETE

Simulated World



Vision-Based Perception

PERCEPTION PIPELINE

RGB-D Camera Capture

640×480 resolution • 90° FOV • Head-mounted

HSV Color Segmentation

Red cup: H[0-8, 172-180] • Green marker: H[40-80]

Contour Detection

Morphological ops • Area filtering • Centroid extraction

3D Position Estimation

Depth lookup • Pixel-to-world transform • Position smoothing

KEY TRANSFORMS

Camera → World Frame

```
x_cam = (u - cx) × depth / f
y_cam = (v - cy) × depth / f
P_world = cam_pos + R_cam × P_cam
```

✓ Hardware-Transferable Design

No ground-truth access for decision making. Robot only knows what cameras see; proper coordinate transforms, realistic perception.

Position Smoothing

10-frame median filter eliminates noise and jitter for stable tracking targets.

Inverse Kinematics Control

JACOBIAN-BASED IK

Damped Least Squares Method

$$\Delta q = J^T (JJ^T + \lambda^2 I)^{(-1)} \times \text{error}$$

Solves for joint velocities given end-effector error. Damping factor λ prevents singularities.

6-DOF Error Tracking

Position
X, Y, Z error to target

Orientation
Gripper pointing towards the cup

7-DOF Arm Configuration

ROBOTIS AI Worker arm with redundancy for obstacle avoidance and smooth trajectories.

STABILITY FEATURES

Velocity Limiting

Per-joint max velocities prevent jerky motion:

[3.0, 3.0, 3.0, 3.0, 2.0, 2.0, 2.0] rad/s

Command Smoothing

Exponential filter blends commands:

$q_new = 0.6 \times q_cmd + 0.4 \times q_prev$

Joint Limit Clamping

All commands respect hardware limits. No oscillation at boundaries.

Result

Smooth, stable motion suitable for carrying liquids without spilling.

Dynamic Hand Tracking & Handover

MEDIAPIPE HAND TRACKING

Real Webcam Input

User's actual hand tracked via laptop webcam. 21 landmarks per hand at 30+ FPS.

Gesture Recognition

Finger angle > 140° → Tracking mode

Finger angle < 90° → Release trigger

Position Smoothing

smoothed_x = 0.4×raw + 0.6×prev

Eliminates jitter while maintaining responsiveness.

DYNAMIC ARM MIRRORING

Core Innovation

Robot arm tracks simulated hand marker in real-time. Human hand → Sim hand → Robot mirrors.

Mirrored Movement

marker_delta = center - marker_x

joint_delta = delta × SCALE × 2.0

Hand right → Arm left (mirrored)

Double Smoothing

Target

$\alpha = 0.7$

Movement

$\alpha = 0.6$

Gesture-Triggered Release

Hold fist for 0.2s → Gripper opens smoothly

System Demo

Complete pick-and-place pipeline with real-time hand tracking and dynamic handover

1
Detect & Pick

2
Transport

3
Track & Handover

▸ Video demonstration follows

Results & Evaluation

SYSTEM CAPABILITIES

Vision-Based Object Detection

RGB-D perception with 3D position estimation. No ground-truth shortcuts.

Coordinated Motion Control

Jacobian IK with velocity limiting. Smooth enough for liquid transport.

Real-Time Hand Tracking

MediaPipe integration with gesture recognition at interactive rates.

Dynamic Handover

Robot adapts to human, not vice versa. Mirrored arm tracking.

TEST SCENARIOS

Scenario 1: Basic Handover

Robot detects cup → picks → transports → user extends hand → release

✓ **Complete pipeline success**

Scenario 2: Moving Target

User moves hand left/right during approach → Robot arm tracks continuously

✓ **Real-time adaptation demonstrated**

Scenario 3: Gesture Control

Palm open → tracking mode • Fist → release trigger

✓ **Intuitive gesture-based interaction**

Challenges & Solutions

CHALLENGE

Arm Oscillation During IK

Arm would oscillate rapidly when approaching target positions.

SOLUTION

Velocity Limiting + Command Smoothing

Per-joint velocity caps and exponential filtering of commands.

CHALLENGE

Arm Snapping at Track Start

Arm would jump to "home" position when hand tracking began.

SOLUTION

Capture Center on First PALM_OPEN

Record marker position when tracking starts, compute delta from there.

CHALLENGE

Cup Slipping During Transport

Cup would slip from gripper during fast base movements.

SOLUTION

Wrist Tilt + Increased Friction

Tilt gripper back 0.4 rad. Patch gripper friction coefficients.

CHALLENGE

Noisy Depth Measurements

Single-pixel depth values caused jittery position estimates.

SOLUTION

Window Median + History Filter

7×7 depth window median, 10-frame position history median.

Conclusion & Future Work

PROJECT SUMMARY

Built a complete vision-based pick-and-place system with dynamic human-robot handover.

- RGB-D perception with proper transforms
- Jacobian-based IK with stability
- Real-time hand tracking & mirroring
- Gesture-triggered release mechanism

Key Achievement

Robot adapts to human, not human to robot. Natural, intuitive interaction.

FUTURE DIRECTIONS

Real Hardware Transfer

Test on physical ROBOTIS AI Worker with calibrated cameras.

Multi-Object Manipulation

Extend to multiple objects with grasp planning.

User Study

Quantitative comparison: dynamic vs static handover preferences.

Learning Component

-> Learn grasp offset from trial and error

-> Imitation Learning for handover trajectory

Thank You

Questions?

Project	Platform	Author
Dynamic Object Handover	**ROBOTIS AI Worker**	**Ansh Bhansali** **anshb3**